

Haar-span density at the limiting Besov edge

A full answer to the finite- q question in arXiv:1901.09117

Solution packet for *Basis properties of the Haar system in limiting Besov spaces*

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Result. If $d/(d+1) \leq p < 1$, then the algebraic span of the inhomogeneous Haar system is dense in $B_{p,q}^1(\mathbb{R}^d)$ if and only if

$$p < q < \infty.$$

Thus every finite case left open in the source has a positive answer.

1 Source question and answer

Garrigós, Seeger and Ullrich prove that $\text{span } \mathcal{H}_d$ is not dense in $B_{p,q}^1(\mathbb{R}^d)$ when

$$\frac{d}{d+1} \leq p < 1, \quad 0 < q \leq p,$$

and ask whether density also fails when $q > p$ [1, §8.1]. They also observe that density fails for $q = \infty$, since the ambient Besov space is nonseparable while the closure of a countable span is separable.

The missing answer goes in the opposite direction from the expectation in the source.

Theorem 1 (Complete density classification). *Let $d \in \mathbb{N}$ and $d/(d+1) \leq p < 1$. Then*

$$\overline{\text{span } \mathcal{H}_d}^{B_{p,q}^1(\mathbb{R}^d)} = B_{p,q}^1(\mathbb{R}^d) \iff p < q < \infty.$$

More quantitatively, if $p < q < \infty$ and $f \in C_c^\infty(\mathbb{R}^d)$, there are finite Haar combinations g_M , indexed by powers of two M , such that

$$\|f - g_M\|_{B_{p,q}^1} \leq C_f (M^{1/q-1/p} + M^{-1}). \quad (1)$$

2 Proof Intuition

A single dyadic conditional expectation $\mathbb{E}_N f$ creates a jump defect of order one in the Besov norm near its natural scale 2^{-N} . Averaging many such expectations globally spreads this defect over scales, but only with cost $M^{1/q-1}$; that proves convergence only for $q > 1$.

The decisive extra freedom is to spread the defects in *space as well as scale*. Divide a cube into M equal thin slabs and use a different dyadic resolution on each slab. A defect occupying a fraction $1/M$ of space has L^p cost $(1/M)^{1/p}$. Put the M slab defects at M different scales and take the ℓ^q scale norm: the total cost becomes

$$(M (M^{-1/p})^q)^{1/q} = M^{1/q-1/p}.$$

$$(57) \quad \|\mathbb{E}_N\|_{b_{\infty,q}^{-1} \rightarrow B_{\infty,1}^{-1}} \gtrsim \|\mathbb{E}_N\|_{b_{1,\infty}^1 \rightarrow B_{1,\infty}^1}.$$

We use that $(b_{\infty,q}^{-1})^* = B_{1,\infty}^1$ for $q \leq 1$; see [15, 2.5.1/Remark 7]. Then for $f \in \mathcal{S}$

$$(58) \quad \|\mathbb{E}_N f\|_{B_{1,\infty}^1} = \|\mathbb{E}_N f\|_{(b_{\infty,q}^{-1})^*} = \sup_{\substack{g \in \mathcal{S} \\ \|g\|_{b_{\infty,q}^{-1}} \leq 1}} |\langle \mathbb{E}_N f, g \rangle|.$$

Now, for each $g \in \mathcal{S}$, since $f = \sum_{j=0}^{\infty} L_j \Lambda_j f$ in $\mathcal{S}$, we have

$$\begin{aligned} |\langle \mathbb{E}_N f, g \rangle| &= |\langle f, \mathbb{E}_N g \rangle| \leq \sum_{j=0}^{\infty} \|\Lambda_j f\|_1 \|L_j(\mathbb{E}_N g)\|_{\infty} \\ &\lesssim \|f\|_{b_{1,\infty}^1} \|\mathbb{E}_N g\|_{B_{\infty,1}^{-1}} \leq \|f\|_{b_{1,\infty}^1} \|\mathbb{E}_N\|_{b_{\infty,q}^{-1} \rightarrow B_{\infty,1}^{-1}} \|g\|_{b_{\infty,q}^{-1}}. \end{aligned}$$

Inserting this into (58) we arrive at

$$\|\mathbb{E}_N f\|_{B_{1,\infty}^1} \leq \|\mathbb{E}_N\|_{b_{\infty,q}^{-1} \rightarrow B_{\infty,1}^{-1}} \|f\|_{b_{1,\infty}^1}$$

and hence (57).

8. DENSITY AND APPROXIMATION

In this section we show two results regarding approximation by linear combinations of Haar functions. The main results in §8.1 are relevant to the formulation of Theorems 1.3 and 1.4 which rule out the case $s = 1$. We shall also obtain a positive result about approximation for the spaces $b_{p,\infty}^{1/p}$.

8.1. *The case $s = 1$.* We shall show that no strongly admissible enumeration of the Haar system can form a Schauder basis on $B_{p,q}^1(\mathbb{R}^d)$ if $\frac{d}{d+1} \leq p < 1$ and $q > 0$. Moreover if in addition $0 < q \leq p$ we shall show that the Haar system is not dense in $B_{p,q}^1(\mathbb{R}^d)$. It seems plausible that this last assertion would continue to hold for all $0 < q \leq \infty$, but we do not have a proof in this generality.

Let us start with an auxiliary result. It is well-known that

$$\|f\|_{B_{p,\infty}^s(\mathbb{R}^d)} \approx \|f\|_p + \sum_{j=1}^d \sup_{|h| < 1} \frac{\|\Delta_{he_j}^2 f\|_p}{|h|^s},$$

Figure 1: The exact open-problem statement on page 28 of the official arXiv:1901.09117v2 PDF.

This tends to zero exactly for $q > p$. The only apparent danger is the new jump along each slab boundary. Choosing all dyadic resolutions sufficiently fine makes those boundary jumps of size 2^{-N_i} ; a geometric local-means estimate shows that their total contribution is $O(2^{-N_0})$.

3 Local-means estimate

We use the compactly supported local-means characterization fixed in the source [1, (12)]. Thus $L_0 u = \beta_0 * u$ and, for $k \geq 1$, $L_k u = \beta_k * u$, where $\beta_k(x) = 2^{kd} \beta(2^k x)$, β has as many vanishing moments as needed, and

$$\|u\|_{B_{p,q}^1} \asymp \left(\sum_{k=0}^{\infty} [2^k \|L_k u\|_p]^q \right)^{1/q}. \quad (2)$$

For $N \geq 0$, let $\mathbb{E}_N$ be conditional expectation on the dyadic cubes of side 2^{-N} .

For $0 < p < 1$, define the summable two-sided sequence

$$W(m) = \begin{cases} 2^{pm}, & m \leq 0, \\ 2^{-(1-p)m}, & m > 0. \end{cases} \quad (m \in \mathbb{Z}). \quad (3)$$

Lemma 1 (A defect confined to one slab). *Fix a compact dyadic cube K and $f \in C_c^\infty(K)$. Let $\Omega \subset K$ be an axis-parallel dyadic rectangle, write $\theta = |\Omega|$ and $S = 1 + \mathcal{H}^{d-1}(\partial\Omega)$, and suppose its boundary is aligned with the level- N dyadic grid. Then, uniformly in k, N and Ω ,*

$$\left[2^k \|L_k(\mathbf{1}_\Omega(f - \mathbb{E}_N f))\|_p\right]^p \leq C_f \left\{ \theta W(k - N) + S, 2^{-pN} 2^{-(1-p)k} \right\}. \quad (4)$$

The constant may depend on K, p, d and finitely many derivatives of f , but not on the thickness of Ω .

Proof. Set $\delta = 2^{-N}$ and, for $k \geq 1$, $t = 2^{-k}$. Since f is globally Lipschitz after extension by zero,

$$|f - \mathbb{E}_N f| \leq C_f \delta. \quad (5)$$

The ct -neighborhood of an axis-parallel rectangle has volume at most $C(\theta + St)$. If $t < \delta$, and G_N denotes the union of the faces of all level- N cubes, elementary counting of the grid faces gives

$$|\Omega^{(ct)} \cap G_N^{(ct)}| \leq C \left(\theta \frac{t}{\delta} + St \right), \quad |(\partial\Omega)^{(ct)}| \leq CSt. \quad (6)$$

First suppose $k \leq N$, so $t \geq \delta$ (with $t = 1$ when $k = 0$). The crude bound (5), compact support of the kernels, and the first tubular-volume estimate imply

$$[t^{-1} \|L_k(\mathbf{1}_\Omega(f - \mathbb{E}_N f))\|_p]^p \leq C_f \delta^p t^{-p} (\theta + St).$$

This is

$$C_f \{ \theta 2^{p(k-N)} + S 2^{-pN} 2^{-(1-p)k} \},$$

which is (4) for $k - N \leq 0$. The same argument applies to the noncancellative low-frequency kernel L_0 .

Now suppose $k > N$, so $t < \delta$. Away from the ct -neighborhoods of G_N and $\partial\Omega$, the convolution kernel stays in one level- N cube and inside Ω . There $\mathbb{E}_N f$ is constant, so cancellation gives

$$L_k(\mathbf{1}_\Omega(f - \mathbb{E}_N f)) = L_k f.$$

If the local-mean kernel has L vanishing moments, Taylor's theorem gives $|L_k f| \leq C_f t^L$. On the two exceptional neighborhoods we use the crude bound $C_f \delta$. Consequently, by (6),

$$\begin{aligned} & [t^{-1} \|L_k(\mathbf{1}_\Omega(f - \mathbb{E}_N f))\|_p]^p \\ & \leq C_f [\theta t^{(L-1)p} + S t^{(L-1)p+1} + \theta \delta^{p-1} t^{1-p} + S \delta^p t^{1-p}]. \end{aligned}$$

Choose L so large that $(L-1)p \geq 1-p$. Because $t \leq \delta \leq 1$, the first two terms are absorbed by the third and fourth, respectively. The remaining main term is

$$\theta \delta^{p-1} t^{1-p} = \theta 2^{-(1-p)(k-N)},$$

and the last term is the stated boundary remainder. This proves the lemma. $\square$

4 Mixed-resolution approximation

Proof of Theorem 1. The negative implications are already in the source: Corollary 8.4 gives non-density for $q \leq p$, and nonseparability gives it for $q = \infty$. It remains to prove density when $p < q < \infty$.

Fix $f \in C_c^\infty(\mathbb{R}^d)$ and choose a dyadic cube K containing its support, with boundary on the unit grid. Let $M = 2^R$ and partition K into M half-open slabs $\Omega_1, \dots, \Omega_M$ of equal width in the first coordinate. Then

$$|\Omega_i| = |K|/M, \quad 1 + \mathcal{H}^{d-1}(\partial\Omega_i) \leq C_K \quad (7)$$

uniformly in i and M .

Choose $N_0 \geq R + C'_K$ so that all slab boundaries lie on the level- N_0 grid, and set $N_i = N_0 + i$. Define

$$g_M = \sum_{i=1}^M \mathbf{1}_{\Omega_i} \mathbb{E}_{N_i} f. \quad (8)$$

Every summand is a compactly supported step function on level- N_i dyadic cubes. By the martingale-difference identity, it is a finite linear combination of inhomogeneous Haar functions; hence $g_M \in \text{span } \mathcal{H}_d$. Moreover,

$$f - g_M = \sum_{i=1}^M u_i, \quad u_i = \mathbf{1}_{\Omega_i} (f - \mathbb{E}_{N_i} f).$$

Since $p < 1$, pointwise p -subadditivity and Lemma 1 give, with $A_k = [2^k \|L_k(f - g_M)\|_p]^p$,

$$A_k \leq C_f \left[\sum_{i=1}^M \frac{|K|}{M} W(k - N_i) + 2^{-(1-p)k} \sum_{i=1}^M 2^{-pN_i} \right]. \quad (9)$$

Let ν be the sequence on $\mathbb{Z}$ having mass $|K|/M$ at each N_i and zero elsewhere. Put $r = q/p > 1$. Since $W \in \ell^1(\mathbb{Z})$, discrete Young's inequality yields

$$\|\nu * W\|_{\ell^r} \leq \|\nu\|_{\ell^r} \|W\|_{\ell^1} \leq C_{p,K} M^{1/r-1}. \quad (10)$$

Also, the levels $N_i = N_0 + i$ make the second sum in (9) at most $C_p 2^{-pN_0}$, and $\{2^{-(1-p)k}\}_{k \geq 0} \in \ell^r$. Combining (2)–(10),

$$\begin{aligned} \|f - g_M\|_{B_{p,q}^1} &\lesssim \|A\|_{\ell^r}^{1/p} \\ &\leq C_f \left(M^{(1/r-1)/p} + 2^{-N_0} \right) \\ &= C_f \left(M^{1/q-1/p} + 2^{-N_0} \right). \end{aligned}$$

Taking $N_0 = R + C'_K$ gives $2^{-N_0} \lesssim_K M^{-1}$ and proves (1). Since $q > p$, the first exponent is negative, so $g_M \rightarrow f$ in $B_{p,q}^1$.

Finally, $C_c^\infty(\mathbb{R}^d)$ is dense in $B_{p,q}^1(\mathbb{R}^d)$ whenever $p, q < \infty$ (see, for example, [3]). Approximating an arbitrary Besov element first by a smooth compactly supported function and then by g_M proves the desired density. $\square$

5 Consequences, checks, and scope

The result completes the density phase diagram in this parameter range:

Range of q	Density of span $\mathcal{H}_d$	Source of conclusion
$0 < q \leq p$	no	Corollary 8.4 of [1]
$p < q < \infty$	yes	Theorem 1 above
$q = \infty$	no	nonseparability, noted in [1]

This is a statement about density only. It does not contradict the source’s proof that no strongly admissible enumeration of the Haar system is a Schauder basis at $s = 1$: dense systems can still have divergent or unbounded ordered partial sums.

The finite script `verify_exponents.py` checks the discrete convolution and the rate $M^{1/q-1/p}$ for representative parameters on both sides of $q = 1$. The geometric proof of Lemma 1, rather than that computation, is the substantive verification.

A bounded search of the run indexes and later arXiv records by exact title, exact question phrase, author names, and core Haar/Besov density terms found no later resolution. The companion Triebel–Lizorkin paper [2] does not transfer the desired conclusion through the standard fixed-smoothness embeddings. The result is therefore presented as a candidate new full solution, subject to expert review.

References

- [1] G. Garrigós, A. Seeger and T. Ullrich, *Basis properties of the Haar system in limiting Besov spaces*, arXiv:1901.09117v2, 2019.
- [2] G. Garrigós, A. Seeger and T. Ullrich, *The Haar system in Triebel–Lizorkin spaces: endpoint results*, arXiv:1907.03738, 2019.
- [3] H. Triebel, *Theory of Function Spaces II*, Monographs in Mathematics 84, Birkhäuser, Basel, 1992.